

Spatio-Temporal Changes of Chlorophyll Concentration with Environmental Parameters Over the Gulf of Mannar, India

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ABSTRACT

The present study investigated the spatio-temporal analysis of chlorophyll concentration with the variation of environmental parameters such as Aerosol optical depth (AOD) and sea surface temperature (SST) over the Gulf of Mannar (GOM) located in the southeastern part of India. The data derived from several satellites already provide information about the aquatic environment throughout the global waters. This study utilized the standard algorithms for the retrieval of *Chl-a* from the Moderate-resolution Imaging Spectroradiometer (MODIS) satellite sensor for a period of about 2002 to 2020. The monthly spatio-temporal variability of the *Chl-a* concentration with the changes in SST and AOD were analyzed over the Gulf of Mannar for the year 2011. This shows that many areas in the Gulf of Mannar have high chlorophyll concentrations of about 2 to 5 mg.m⁻³ in the single-day images for several months during 2011. Also, the variation in the SST is observed for several months during 2011, ranging from 28 to 31.5°C. The AOD values fall in the range of 0.09 to 0.12. In addition, the time series analysis of *Chl-a*, SST, and AOD were also studied for two different stations over the GOM. The result of the time series analysis explains that high chlorophyll concentration was observed during the colder months with low values in SST and AOD.

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INTRODUCTION

Chlorophyll-a (*Chl-a*) is an important water quality index in both marine and coastal environments. *Chl-a* in the ocean is the indicator of phytoplankton biomass (Gitelson et al. 1993) and primary productivity (Cannizzaro & Carder 2006, Huot et al. 2007). Phytoplankton are microscopic plants that form the base of the marine food web, and their growth and distribution are influenced by various environmental factors. *Chl-a* is the most abundant pigment in phytoplankton (Patissier et al. 2014). *Chl-a* is the biological component accessible to remote sensing (via ocean color) and provides a key metric for assessing the health and productivity of marine ecosystems globally (Gurlin et al. 2011). *Chl-a* absorbs sunlight and carbon dioxide from the earth and supports the primary production (Hu et al. 2012), which is good for overall ocean health. It is responsible for the appearance of green pigments in plants and algae (Platt & Sathyendranath 2008). When the chlorophyll concentration increases, the water color shifts slightly varies from blue to green. This indicates an increase in the phytoplankton population (Simon & Shanmugam 2012). Regular monitoring is necessary for assessing the blooms and implementing effective management strategies to avoid the deterioration of water quality (Le et al. 2013).

The extension of three different one-dimensional OCDMA codes into a three-dimensional OCDMA system for improved performance. The three distinct codes are the non-mapping, MS, and MD. The extensions are such that the spectral domain utilizes the non-mapping code, time spreading uses the



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MS code, and spatial domains extend with the MD code (Gupta et al. 2024).

The growth of phytoplankton is often limited by the availability of nutrients such as nitrogen, phosphorus, and silica. Areas with upwelling or high nutrient input typically have higher chlorophyll concentrations. The increase in nutrient supply primarily increases the phytoplankton growth rate (Banse et al. 1996). The *Chl-a* concentration is mainly affected by environmental and climatic factors such as rainfall, currents, seasonal atmospheric cycles, inland river water inflow, local wind regimes, water column stratification, and sea surface temperature (*SST*) volatilities (Tian et al. 2016).

Sea Surface Temperature (*SST*) is the important parameter that affects the phytoplankton growth rates and community composition. Extreme temperatures can inhibit phytoplankton growth or lead to harmful algal blooms. Variations in *SST* can affect the marine ecosystem in various ways (Brochier et al. 2013). According to the Intergovernmental Panel on Climate Change (IPCC), global sea surface temperature is expected to rise by approximately 0.4 to 1.1°C by 2025. Change in surface temperature affects the plant and animal species present in the ocean. An increase in *SST* may decrease the *Chl-a* concentration and affect the circulation pattern that brings nutrients from the deep sea to surface water. An excessive increase in *SST* destroys the growth of phytoplankton, and another factor that affects the growth is the amount of nutrients loaded with freshwater from river discharge (Nuridin et al. 2013). Changes in sea surface temperature and *Chl-a* concentration may lead to global warming (Gunasinghe et al. 2024, Mondal & Lee 2024).

The *Chl-a* concentration can also be affected when the change in salinity occurs due to the evaporation of seawater (Al-Tae 2018). When the wind speed increases, the evaporation rate also increases, which causes a change in salinity and *Chl-a* concentration (Mahpeykar & Khalilabadi 2021). Aerosols are rich sources of nutrients that increase the production of marine environment and their carbon sequence (Paytan et al. 2009). Aerosols contain a variety of natural and synthetic compounds, including mineral dust, sea salt crystals, bacteria, and other microscopic particles. Aerosols are solid or liquid particles suspended in the atmosphere with a diameter of approximately 0.001 to 100 µm (Giles et al. 2019). Aerosol particles not only affect the radiative forcing of the Earth's air system but also influence climate and cause environmental problems such as acid rain and haze (Li et al. 2018, Huang & Ding 2021). High concentrations of aerosols cause serious threats to human health (Morozova & Mironova 2015). The properties of aerosols include parameters such as aerosol optical depth (AOD), single-

scattering albedo, scattering phase function, and absorbing optical depth. AOD is the most important parameter, which is defined as the integral of the aerosol extinction coefficient in the vertical direction. This describes the attenuation effect of aerosols on light and is an important indicator of the level of air pollution.

Estimating *Chl-a* concentration using the remote sensing data is challenging in turbid and productive waters. It is difficult to monitor the changes in the *Chl-a* concentrations through field measurement and cost high. This can be overcome by using satellite measurements through remote sensing techniques, which are of low cost with high efficiency. Several studies have been conducted to monitor the water quality on a large scale using remote sensors (Le et al. 2013, Tong et al. 2022, Koponen et al. 2007). Satellite sensors provide reliable global ocean coverage of *Chl-a* and *SST* at relatively high spatial and temporal resolution. The spatial and temporal distribution can be effectively measured from space (Ishizaka 2010).

Several satellites like Coastal Zone Color Scanner (CZCS), Sea-viewing Wide Field-of-View sensor (SeaWiFS), Moderate Resolution Imaging Spectrometer (MODIS), Medium Resolution Imaging Spectrometer (MERIS) and VIIRS, etc. are used to derive *Chl-a* products from the ocean and coastal waters. NASA CZCS, launched in 1978, is designed to monitor ocean color. Ocean and Land Color Instrument (OLCI) onboard the sentinel-3 satellite is introduced to monitor water quality in turbid lakes. OLCI satellite includes 21 spectral bands, improved signal-to-noise ratio (SNR), cameras for mitigation of sun glint, a global spatial resolution of 300 m, daily revisit times, and enhanced coverage. Several research has attempted to map *Chl-a* concentration based on OLCI images in various turbid lakes (Pahlevan et al. 2020, Shen et al. 2020).

Satellite remote sensing is the best way to view the entire ocean surface and to monitor *Chl-a* at high spatial and temporal resolution (Brewin et al. 2021). To derive *Chl-a*, remote sensing has employed empirical algorithms (Dierssen 2010) and semi-analytical algorithms (IOCCG 2006). The empirical algorithm, known as the ocean color algorithm, uses absorption properties of *Chl-a* and remote sensing reflectance to eliminate *Chl-a* from space (O'Reilly & Werdell 2019, O'Reilly et al. 1998). The semi-empirical algorithm uses physical models of light interactions with water to eliminate the *Chl-a* (Werdell et al. 2018).

Remote sensing reflectance (*R_{rs}*) is determined from the inherent optical properties (IOPs) of water and optically active water constituents (OAC) (Gordon et al. 1988, Mobley 1999). Estimation of *Chl-a* from *R_{rs}* can be analytically approached by forward or inverse modeling or by empirically

associated variability of Rrs with one or more wavebands (Schalles 2006). When the phytoplankton increases in the ocean, the shift of reflectance is observed from blue to green wavelengths caused by the combination of increased absorption and particulate backscattering process (Morel & Prieur 1977). This observation is then converted to the most widely used Ocean Color (OC) algorithm, in which several variations occur for specific sensors (O'Reilly et al. 1998, O'Reilly & Werdell 2019). The empirical algorithms are specially designed to estimate *Chl-a* from waters with high algal biomass and identify algal blooms based on peak height methods (Matthews et al. 2012).

Several studies have been conducted to understand the variability of *Chl-a* in coastal waters using satellite sensors. Chen et al. (2013) presented *Chl-a* concentration retrieval in the coastal waters of Bohai, Yellow, and East China Sea. The in-situ data were collected and are used to assess the performance of HY-1C in retrieving the *Chl-a*. The result of this study shows that the algorithm based on blue-green band ratios performed well, and the broadband channel of HY-1C CZ1 can retrieve *Chl-a* in turbid waters. Alexander et al. 2010 made a comprehensive study on the algorithm for retrieving the *Chl-a* from the coastal waters. The two and three-band algorithms are tested with the bands that matched the spectral channels of MERIS.

In this study, the changes in *Chl-a* concentrations were analyzed with *SST* and *AOT* using MODIS satellite images. The study is conducted in the Gulf of Mannar, where the coastal waters are dominated by Chlorophyll-a and are

monitored using satellite sensors. Here, the OC3 algorithm is used for the retrieval of *Chl-a* from Rrs. The analysis includes the spatio-temporal analysis of *Chl-a* with *AOT* and *SST* variation using the monthly images over the year 2011. Further, the study extended to the time series analysis of *Chl-a* with the changes of environmental parameters such as *AOD* and *SST* analyzed over the years 2000-2020.

STUDY AREA

Gulf of Mannar (GOM) (8.47°N, 79.02°E) is located in the Indian Ocean between Southeastern India and Western Sri Lanka. GOM extends from Tuticorin to Rameswaram Island in the SE-NE direction. There are 21 islands situated at an average distance of about 8 km from the coast and running almost parallel to the coastal line. GOM has a rich and diverse fauna compared to other regions. GOM region is enriched with various habitats such as Coral reefs, mangroves, sea grasses, seaweeds, etc. GOM is a biologically rich area and is identified with more than 4223 species of flora and fauna. The Gulf of Mannar typically exhibits moderate to high levels of chlorophyll-a present in the nearshore waters along the Indian coast.

The stations used for the analysis are shown in Fig. 1. The study area of station 1, with a latitude of 9.7138 and longitude of 79.1568, is located near the coastal region, which is shallow water, and station 2, with a latitude of 9.3199 and longitude of 79.2026, is located near the Pamban bridge. In the Gulf of Mannar, the water temperature is influenced by the atmospheric temperature. From January to April, the

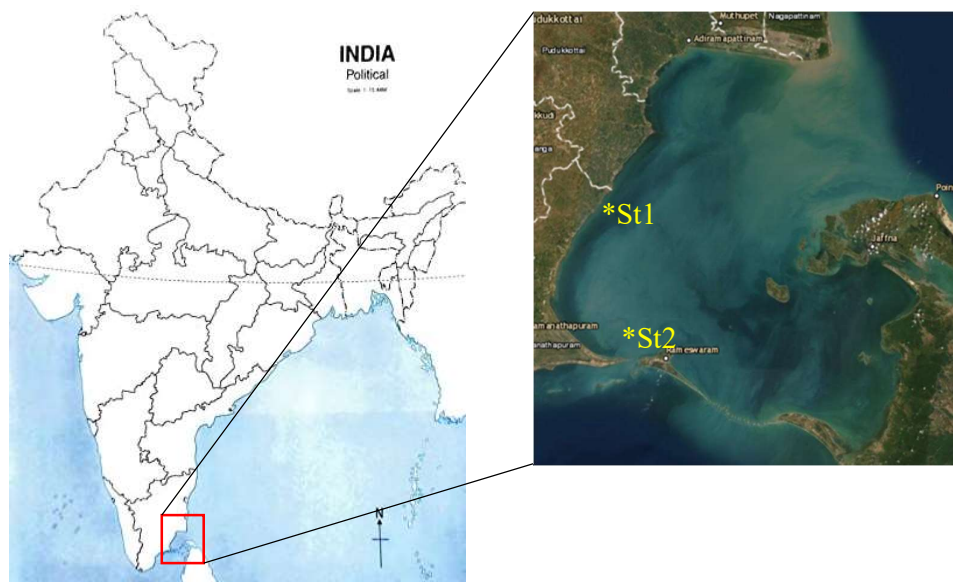


Fig.1: The location of the study area over the Gulf of Mannar in the southeast coastal waters of India (Left side). The data from stations st-1 and st-2 are used for the time series analysis (right side).

temperature steadily increases, with a highest peak of 32°C in May. And till August, the temperature gradually decreases. Between September and October, the temperature again increases and falls between December and January. The water temperature may exceed the atmospheric temperature during summer in shallow regions closer to land.

In the Gulf of Mannar, the *Chl-a* concentration is high during the northeast monsoon. Because, cold water has higher *Chl-a* than warm water, which contains nutrients that come from the deep ocean (Balasubramanian & Shanmugam 2015). *SST* are mostly cooler than the nearby waters where the ocean current causes upwelling, and the *Chl-a* concentration is high. The Gulf of Mannar, being a tropical region, has a warm climate with low change in surface temperature (Tholkapiyan 2012). It is easier for the deep water to rise to the surface when the surface water is cold.

MODIS Aqua satellite data is used for accurately mapping the *Chl-a* concentration in the coastal water of the Gulf of Mannar. MODIS is a satellite-based sensor used to measure climatic conditions. Terra (EOS AM, launched in 1999) and Aqua (EOS PM, launched in 2000) satellites are the two MODIS sensors. MODIS sensors can capture data in 36 spectral bands ranging in wavelength from 0.4 μm to 14.4 μm and at varying spatial resolutions. The captured data from the top of the atmosphere provides Rrs after undergoing atmospheric correction. This Rrs dataset is used to monitor the measurements in large-scale global dynamics, including piles of earth cloud cover and processes occurring in the

ocean, land, and the lower atmosphere. MODIS sensor plays a major role in predicting global changes accurately.

Using the MODIS Aqua sensor, the daily images for the two stations of the Gulf of Mannar are collected over the year 2000 to 2020 from the Ocean Color website. The collected data is then processed, and atmospheric correction is carried out using the ACOLITE software (Pahlevan et al. 2021). The ACOLITE processor was used to correct the atmospheric effects using the dark spectrum fitting (DSF) (proposed by Vanhellemont & Ruddick 2018, Vanhellemont 2019) over SWIR-based exponential extrapolation. Sun glint effect can also be removed using the ACOLITE processor. Using the satellite images, the *Chl-a* is estimated with good spatial resolutions.

MATERIALS AND METHODS

In this study, *Chl-a* concentration was analyzed with *SST* and *AOD* using the Moderate-resolution imaging spectroradiometer (MODIS) satellite sensor over a 20 years from 2000-2020. The standard atmospheric correction procedure (Bailey et al. 2009) is used to perform the atmospheric correction process in which the top-of-atmosphere radiance received by sensors is converted to surface reflectance. OC3 Chlorophyll algorithm is used to retrieve the *Chl-a* from the remote sensing reflectance (R_{rs}), which is developed based on the semi-analytical and bio-optical remote sensing reflectance, R_{rs} . The sea surface temperature (*SST*) measurement and Aerosol optical

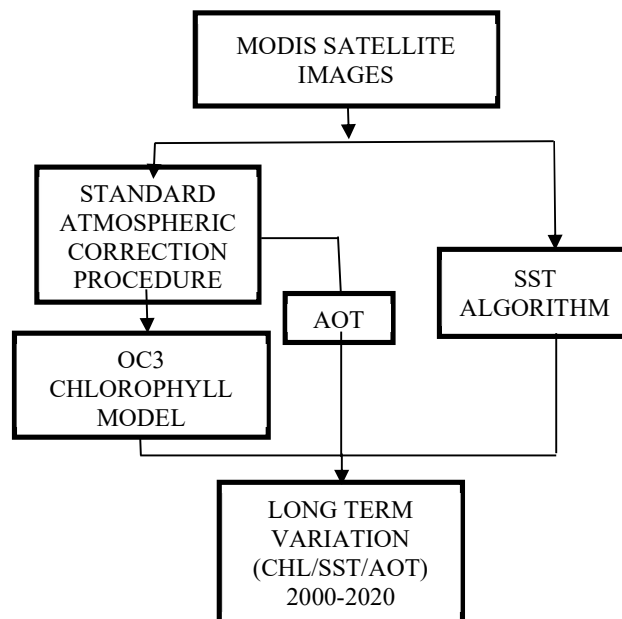


Fig. 2: The schematic block diagram represents the methodology for the estimation of *Chl-a* concentration, *AOT*, and *SST* for understanding their long-term changes by processing the MODIS satellite sensor data from 2000-2020.

depth (AOD) are also derived for this analysis from the MODIS satellite sensor. The variation in the chlorophyll concentration, Aerosol optical depth measurement, and surface temperature analysis is presented in this study. Fig. 2 represents the schematic diagram of the methodology.

Atmospheric correction is a crucial step in remote sensing for accurately estimating surface properties like chlorophyll-a (*Chl-a*) concentrations from satellite data (Gordon et al. 1998). Atmospheric correction aims to remove or correct the effects of the atmosphere on the observed radiance, which includes scattering and absorption of light by atmospheric particles and gases. The atmospheric correction algorithm is the estimation of spectral aerosol reflectance $\rho_a(\lambda)$, subtracted from total measured signals $R_{rs}(\lambda)$. NASA Ocean Biology Processing Group (OBPG) implemented an algorithm that (Gordon & Wang 1994) requires an assumption of negligible water-leaving reflectance in the NIR of the spectrum (i.e., $R_{rs}(\text{NIR}) = 0 \text{ sr}^{-1}$).

Ocean Color remote sensing involves the estimation of concentration in the ocean by measuring the variation in the spectral quality of the water surface. OC is obtained from the incident light interactions with other substances present in water (*Chl-a* concentration, suspended sediments, and CDOM). These substances are quantified by water leaving radiance, which is measured in the visible portion of electromagnetic radiation. OC algorithm is the band-ratio-based algorithm used for the retrieval of *Chl-a* from the remote sensing reflectance, R_{rs} . In the empirical OC algorithm, *Chl-a* has log-normal distribution, which means the model is trained on log-transformed (with base 10) in situ measured *Chl-a*. OC3 algorithm uses a quadratic polynomial with a single explanatory variable. For the OC3 algorithm, the numerator is the maximum of the bands $R_{rs} 443$ and $R_{rs} 488$, and the denominator with $R_{rs} 551$. The OC3 algorithm uses a three-band blue-green reflectance ratio is defined as:

$$R = \log_{10} \left[\frac{R_{rs443} > R_{rs488}}{R_{rs551}} \right] \quad \dots(1)$$

$$Chl-a = 10^{0.28 - 2.753R + 1.457 R^2 + 0.659 R^4}$$

where, *Chl-a* is Chlorophyll-a concentration $\text{mg} \cdot \text{m}^{-3}$, R is Blue-green band ratio. OC-3 algorithm has been used to retrieve low as well as high *Chl-a* concentrations and is also useful in monitoring the waters from estuarine regions using spatial resolution sensors (Morel & Maritorena 2001).

SST maps are based on observations by the MODIS Aqua satellite. The temperature estimated from the satellite measurements is known as brightness temperature (BT). *SST* measured at the sea surface is not the same as the satellite-derived *SST* because of atmospheric interference. So, for the identification or removal of clouds and aerosols' contribution to satellite radiance, an atmospheric correction procedure

was followed. The bands that provide more information for the correction procedure are the atmospheric windows at the wavelength range of $\lambda = 3.5\text{--}4.1 \mu\text{m}$ and $\lambda = 9.5\text{--}12.5 \mu\text{m}$ (McMillin 1975). Although the two spectral channels provide BT, the magnitudes of the retrievals are different based on the atmospheric transmissivity. The difference in brightness temperature is defined as the temperature drop of *SST* and other BT (McMillin 1975, Barton 1995), which provides the linearized relation in terms of the multi-channel sea surface temperature equation. The equation is as follows:

$$SST = aT_i + \gamma(T_i - T_j) + c \quad \dots(2)$$

where γ is the coefficient of differential absorption, T_i and T_j are the brightness temperature obtained from the two channels. MODIS Aqua/Terra has two bands in the transmission window. The robust algorithm is as follows:

$$SST = a_0 + a_1 T_{3.95} + a_2 (T_{3.95} - T_{4.05}) + a_3 (\sec(\theta) - 1) \quad \dots(3)$$

Here, the coefficients of the equation, such as a_0 , a_1 , a_2 , and a_3 , are obtained from Kilpatrick et al. (2015). The above method has been applied to produce satellite-derived *SST* for the temporal and spatial analysis and the *SST* retrieval from the MODIS images.

RESULTS AND DISCUSSION

This section describes the results in two subsections: one is the spatio-temporal analysis, and the second one is the time series analysis. The spatio-temporal analysis of *Chl-a* concentration, AOD, and *SST* are presented for one year (2011) with monthly images. The time series analysis is carried out to investigate the possibility of a significant trend in the *Chl*, *SST*, and AOD over 19 years.

Spatiotemporal Analysis

The above figures represent the *Chl-a*, *AOT*, and *SST* measurements over the Gulf of Mannar by MODIS satellite sensor for the 6-month duration from January to June 2011. The variability of *Chl-a* concentration with the changes in *SST* and *AOT* over the Gulf of Mannar is presented in this study. Fig. 3a. Represents the *Chl-a* concentration, and the color bar represents the relative changes corresponding to the dimensional trend for the *Chl-a* concentration. Fig. 3b represents the Aerosol optical depth measurement, and the color bar indicates the relative change for the aerosol measurement. Fig. 3c represents the surface temperature, and the color bar indicates the relative changes with respect to sea surface temperature.

The *Chl-a*, *AOT*, and *SST* measurements are obtained from the global MODIS ocean color satellite. Data collection

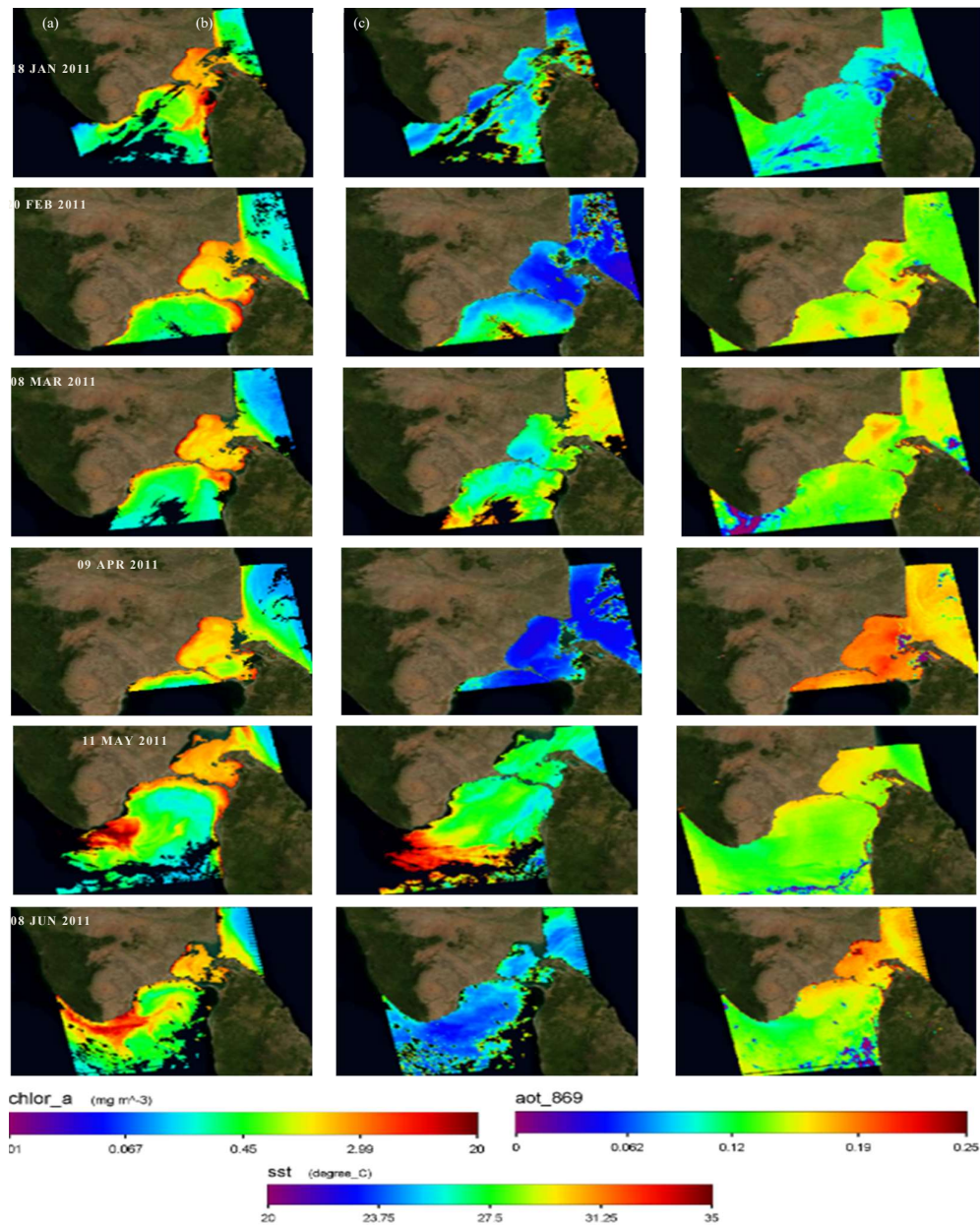


Fig. 3: (a) *Chl-a* concentration trend over the Gulf of Mannar by using a MODIS satellite sensor for the duration of 6 months from January to June 2011, (b) Aerosol optical depth, and (c) Sea surface temperature measurement during the study period. The color bar scale represents the relative changes corresponding to the dimensional trend for the *Chl-a* concentration, AOT, and SST values.

from satellites involves frequent measurements over time with large spatial coverage, which is useful for assessing long-term changes. The *Chl-a* concentration in the Gulf of Mannar shows different seasonal variability for different regions. The *Chl-a* concentration and SST are highly dependent due to the difference in gaining the heat between shallow and deeper portions of the sea. The deeper sea observes high temperatures. The surface water in the middle

region gains a lower temperature than the water near the shore. An increase in surface temperature in shallow water increases the concentration of *Chl-a*, while an increase in surface temperature in the deeper water tends to decrease the concentration of *Chl-a*. During summer, in shallow regions closer to land, the water temperature may exceed even the atmospheric temperature. GOM is characterized by high salinity and surface temperature during the northeast

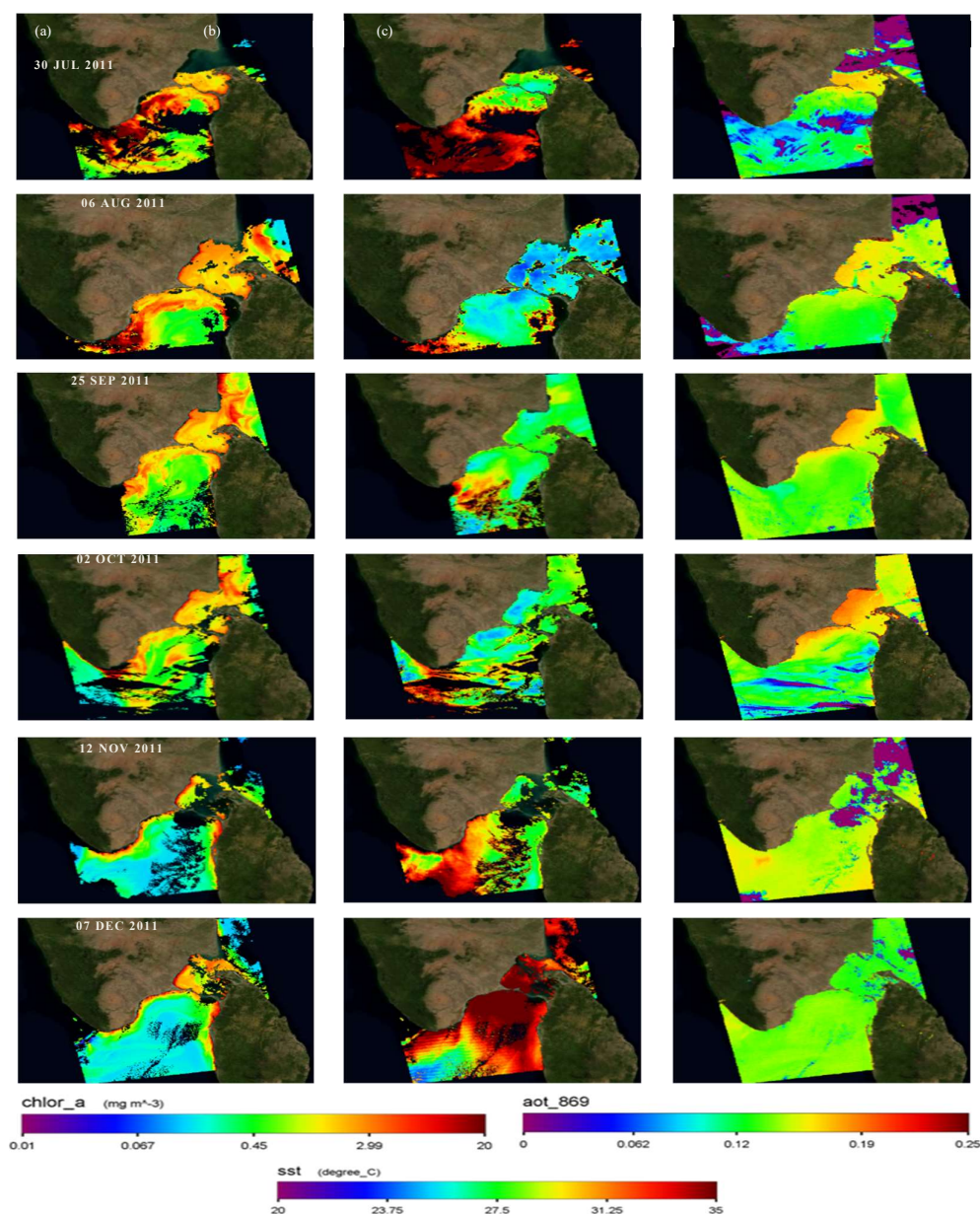


Fig. 4: (a) *Chl-a* concentration trend over the Gulf of Mannar by using a MODIS satellite sensor for the duration of 6 months from July to December 2011, (b) Aerosol optical depth measurement during the study period, and (c) Sea surface temperature measurement during the study period. The color bar scale represents the relative changes corresponding to the dimensional trend for the *Chl-a* concentration, AOT, and SST values.

monsoon. When more heat is absorbed in the ocean, surface temperature increases, and the ocean circulation patterns transport warm and cold water around the globe. From Fig. 3a, very low numbers of chlorophyll concentrations appear in light green, and high chlorophyll concentrations appear in dark orange. The observations were taken from the MODIS sensor on NASA's Aqua satellite. The satellites measure the temperature of the top millimeter of the ocean surface. From Fig. 3c, the coolest water temperature appears

in blue, and the warmest temperature appears in red. In the Gulf of Mannar, the Northeast Monsoon is steadiest in January and is weak by March. The wind becomes light and variable toward the end of April.

The northern part of GOM experiences high surface pigment concentration. The waters are very shallow compared to other surrounding waters of the island. Thus, the water contains other substances like suspended particles and dissolved organic matter in addition to *Chl-a*. From the

above figure, *Chl-a* concentration is high during January which is due to the decrease in the *AOT* and the *SST*. *SST* is high during April. Thus, the *Chl-a* concentration and the *AOT* measurements are reduced due to the high surface temperature. During the months of March to May, the *Chl-a* concentration was reduced when compared with January, which may be due to surface heating during summer. From February to May, there was minimal variation in chlorophyll concentration. During June, the *Chl-a* concentration is low which is due to the increase in the surface temperature. The highest spatially mean *Chl-a* concentration is observed in January, and the lowest is observed in June.

Spatial and temporal analysis shows that *SST* and *Chl-a* concentration in this area is greatly influenced by the monsoon. *Chl-a* is an indicator of photosynthesis present in the ocean. *Chl-a* concentration is influenced by climatic factors such as surface temperature and winds. In GOM, *Chl-a* concentration is high during the Northwest monsoon compared to the Southwest monsoon. The *Chl-a* concentration is high along the coast due to the nutrient-rich run-off from land via rivers. Also in near shore, *Chl-a* variability was high and decreases with distance towards offshore. *Chl-a* concentration is affected by rainfall and

river discharge. During the southwest monsoon, *SST* is low compared to the Northwest monsoon, while *Chl-a* in the Northwest monsoon is higher than the Southwest monsoon. After the SW monsoon, the southwest coasts have higher surface pigment concentrations. In the SW monsoon, satellite data were not available, in which these waters are rich in *Chl-a* concentration. During the Southwest monsoon period from July to September, high wind speed and a decrease in surface temperature are observed around the southeastern coast. Hence increase in chlorophyll concentration is observed from August to October 2011.

Fig. 4. represents the *Chl-a*, *AOT*, and the *SST* measurements over the Gulf of Mannar by MODIS satellite sensor for the 6-month duration from July to December 2011. The color bar scale represents the relative changes corresponding to the dimensional trend for the *Chl-a* concentration, Aerosol depth measurement, and surface temperature values. During July, the surface temperature measurement was very low. So, the reduction in the *Chl-a* concentration and the aerosol optical depth measurement is observed during July which is due to the decrease in the surface temperature level. Fig. 4a indicates that the *Chl-a* concentration is high during August and September. In

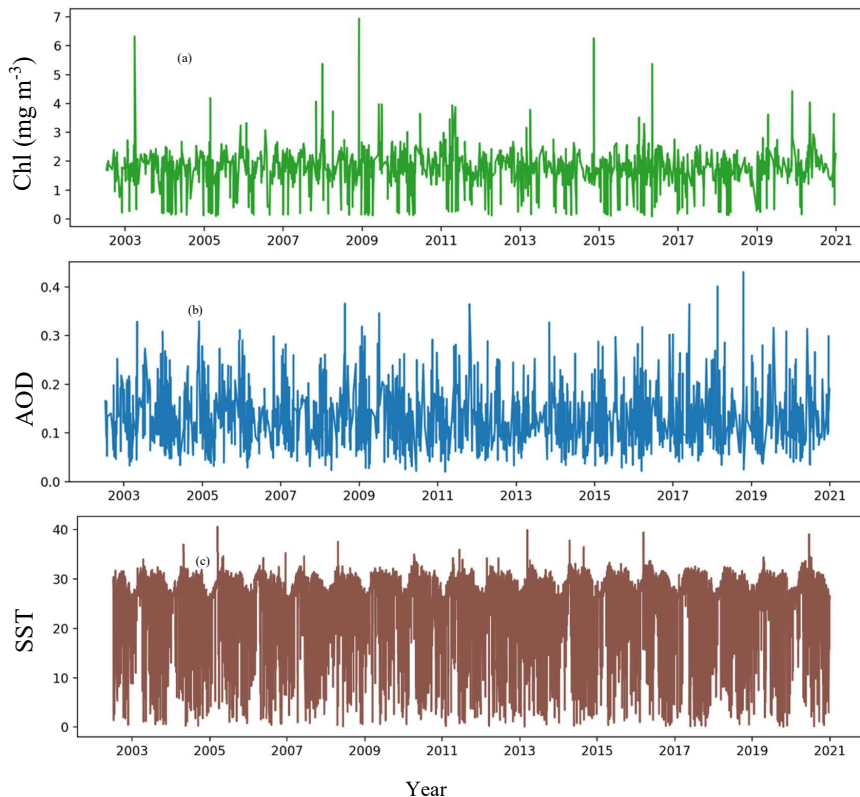


Fig. 5: Variation of *Chl-a*, *AOD*, and *SST* from MODIS data for location (st-1) over the Gulf of Mannar from 2002 to 2020. (a) Time series representation of *Chl* data (b) time series representation of *AOD* data (c) Time series representation of *SST* data.

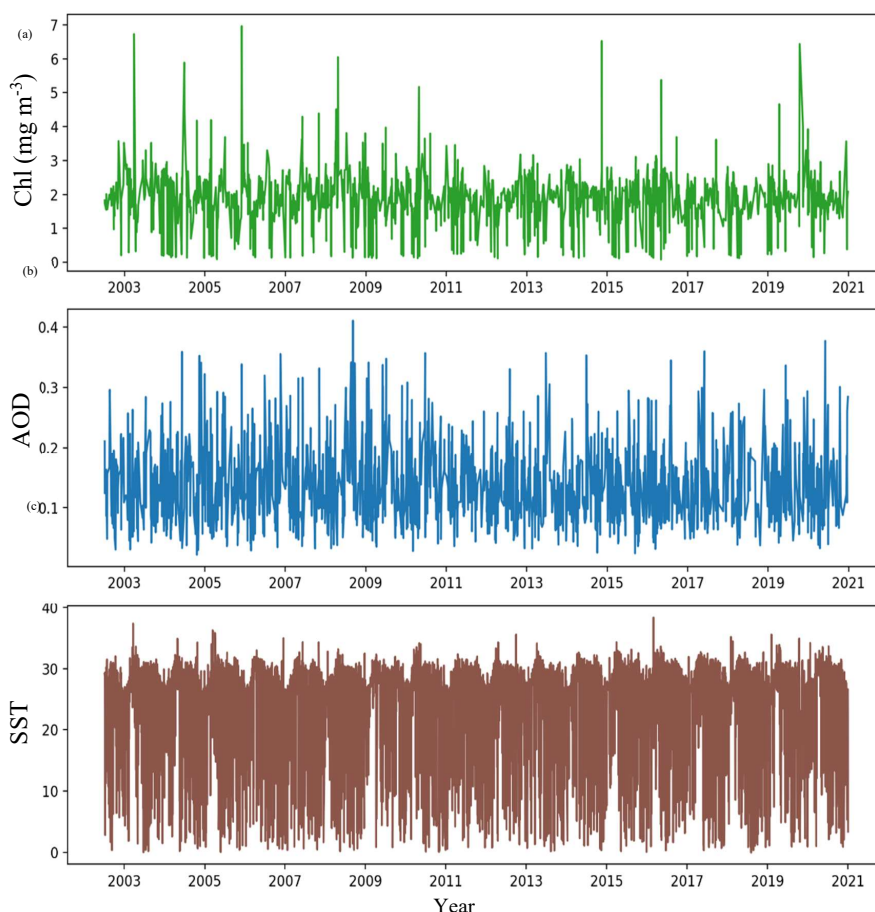


Fig. 6: Variation of *Chl*, *AOD*, and *SST* from MODIS data for location (st-2) over the Gulf of Mannar from 2002 to 2020. (a) Time series representation of *Chl* data (b) time series representation of *AOD* data (c) Time series representation of *SST* data.

August due to high *Chl-a* concentration, the aerosol optical depth measurement is low and an average range of surface temperature measurement. High *Chl-a* concentration in September shows an average range of *AOT* and surface temperature measurements. Increased levels of *Chl-a* concentration and the aerosol optical depth measurement observed during December experienced average surface temperature. During the month of August-October, the southwest monsoon period, results in a high *Chl-a* concentration, which shows the low range of *AOT* and average surface temperature measurements. High *Chl-a* concentration in the open ocean waters is observed due to upwelling rather than other particles. From Fig. 4b very high level of optical depth measurement is obtained during December. So, due to increased aerosol measurement, the *Chl-a* concentration increases with low surface temperature.

Long-Term Time Series Analysis

The time series distribution of *Chl-a* concentration, *AOD*, and *SST* data are obtained from MODIS sensors for the

daily data from 2002 to 2020. The time series distribution of the long-term changes in *Chl-a* concentration was unevenly distributed over the years in station 1. The *Chl-a* concentration was high over station 1, which may be due to the upwelling in the coastal region. Over the study period of about 19 years, monthly *Chl-a* showed a range of 1.8 to 2.7 mg.m^{-3} during the cooler months of January and February and about 1.5 to 2 mg.m^{-3} during the warmer months of July and August, indicating higher *Chl-a* concentrations during the latter period. This is likely due to coastal waters being calmer during the summer than in winter. Throughout the years, during warmer months, the *Chl-a* concentration is about 2 mg.m^{-3} , ranging from 1.8 to 2.2 mg.m^{-3} , and during the colder months, the average *Chl* concentration of about 2.4 mg.m^{-3} , ranging from 2 to 2.7 mg.m^{-3} , was observed.

The time series distribution of *SST* increases during the warmer months and decreases during colder months which may be due to the change in climatic conditions, as shown in Figure 5c. The annual average *SST* was observed during 2011 at about 28.3°C. The overall analysis shows that GOM

experiences an average *SST* of about 27.5°C. *SST* shows a minimum in July and reaches a maximum in April. The time series distribution shows that the *AOT* values increase in linear were observed during the beginning of every year and gradually decrease to minimum values at the end of the year. The average range of *AOT* is about 0.15. High nutrients are observed during summer in which the temperature recorded is also high. The *Chl-a* concentration values are observed as high during the winter season.

The time series distribution of *Chl-a* concentration, *AOD*, and *SST* data for station 2 is obtained from MODIS sensors for the daily data from 2002 to 2020 (Fig. 5). The time series distribution of the long-term changes in *Chl-a* concentration, *AOD*, and *SST* were unevenly distributed over the years in station-2. This station is located near the Pamban bridge, in which the moving of ships occurs, which will affect the *Chl-a* concentration. The *Chl-a* concentration was low in this area which is due to several disturbances that occur in the water. In the western region, *chlorophyll-a* tends to be low, which is due to the strong tidal mixing that causes the residence time of algae in the photic zone. Whereas in the east region, *chlorophyll-a* will be higher. Monthly average *Chl-a* concentration ranges from 1.5 to 2.2 mg.m⁻³. The distribution of *SST* in the east begins to rise in November to April, with an increase in March, and decreases in May to October, with the lowest *SST* in August. The monthly long-term variation for the *SST* from 2002 to 2020 shows that high surface temperature with an average of about 30°C (Fig. 6). Due to high sea surface temperature the *Chl-a* concentration for this area gets reduced. The monthly variation data of *AOT* shows an average of about 0.15 to 0.18.

CONCLUSIONS

Understanding the variations in *Chl-a*, *SST*, and *AOD* in the GoM is crucial, as these parameters significantly influence marine ecosystems and living organisms. The GoM's environmental conditions are particularly unique due to the presence of coral reefs, seagrass beds, and mangroves, which serve as essential habitats for various marine species by providing spawning, feeding grounds, and shelter. This study examines the variability of *Chl-a* in relation to environmental parameters using a satellite-based correction procedure for estimating *Chl-a* concentration, bypassing the need for additional validation steps with in-situ data, as previous research has established the accuracy of satellite-derived measurements. By analysing MODIS satellite data from 2011, we explored the spatio-temporal variability of *Chl-a*, *AOD*, and *SST* across the GoM. Our results indicate that *Chl-a* concentrations are generally lower in coastal regions than in open waters. Additionally, a time-series analysis

of *Chl-a* from 2002 to 2020 revealed correlations between *Chl-a*, *AOD*, and *SST*, showing that increases in *SST* and *AOD* tend to coincide with decreases in *Chl-a* and vice versa. Seasonal fluctuations in *SST*, *AOD*, and *Chl-a* concentrations further suggest that changes in seawater temperature and salinity play a significant role in regulating *Chl-a* levels.

The study also concludes that *Chl-a* concentration decreases with increasing water depth and distance from the shoreline and that *Chl-a* exhibits a strong correlation with physical parameters such as *SST* and salinity. The findings of this study provide valuable insights for the remote sensing community, contributing to the understanding of marine ecosystems and can aid in identifying potential fishing zones. Future work should focus on expanding the spatial coverage of this research and incorporating in-situ or ground truth data to validate satellite observations in other regions where satellite data may be limited.

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